

Assignment 5: Data Visualization

Mickey Zhang

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

#1

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /home/guest/R/EDE_Fall2025/EDE_Fall2025
```

```
library(cowplot)
```

```
##  
## Attaching package: 'cowplot'  
##  
## The following object is masked from 'package:lubridate':  
##  
## stamp
```

```
library(lubridate)  
here()
```

```
## [1] "/home/guest/R/EDE_Fall2025/EDE_Fall2025"
```

```
CN = read.csv(  
  file = here('Data', 'Processed_KEY',  
              'NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv'),  
  stringsAsFactors = TRUE  
)  
LMT = read.csv(  
  file = here('Data', 'Processed_KEY',  
              'NEON_NIWO_Litter_mass_trap_Processed.csv'),  
  stringsAsFactors = TRUE  
)  
#2  
class(CN$sampleddate)
```

```
## [1] "factor"
```

```
CN$sampleddate = as.Date(CN$sampleddate)  
class(CN$sampleddate)
```

```
## [1] "Date"
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3
mytheme = theme_bw()+
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    legend.position = "bottom"
  )
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (tp_ug) by phosphate (po4), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
#4
ggplot(CN, aes(x = po4, y = tp_ug, color = lakename))+
  geom_point(size = 1) +
  geom_smooth(method = "lm", se = FALSE)+
  scale_x_continuous(expand = c(0, 0), limits = c(0, 50)) +
  scale_y_continuous(expand = c(0, 0), limits = c(0, 150)) +
  labs(
    title = "Total Phosphorus vs. Phosphate",
    subtitle = "for Peter and Paul Lakes",
    x = "Phosphate (P04)",
    y = "Total Phosphorus Concentration(tp_ug)",
    color = "Lake"
  ) +
  mytheme
```

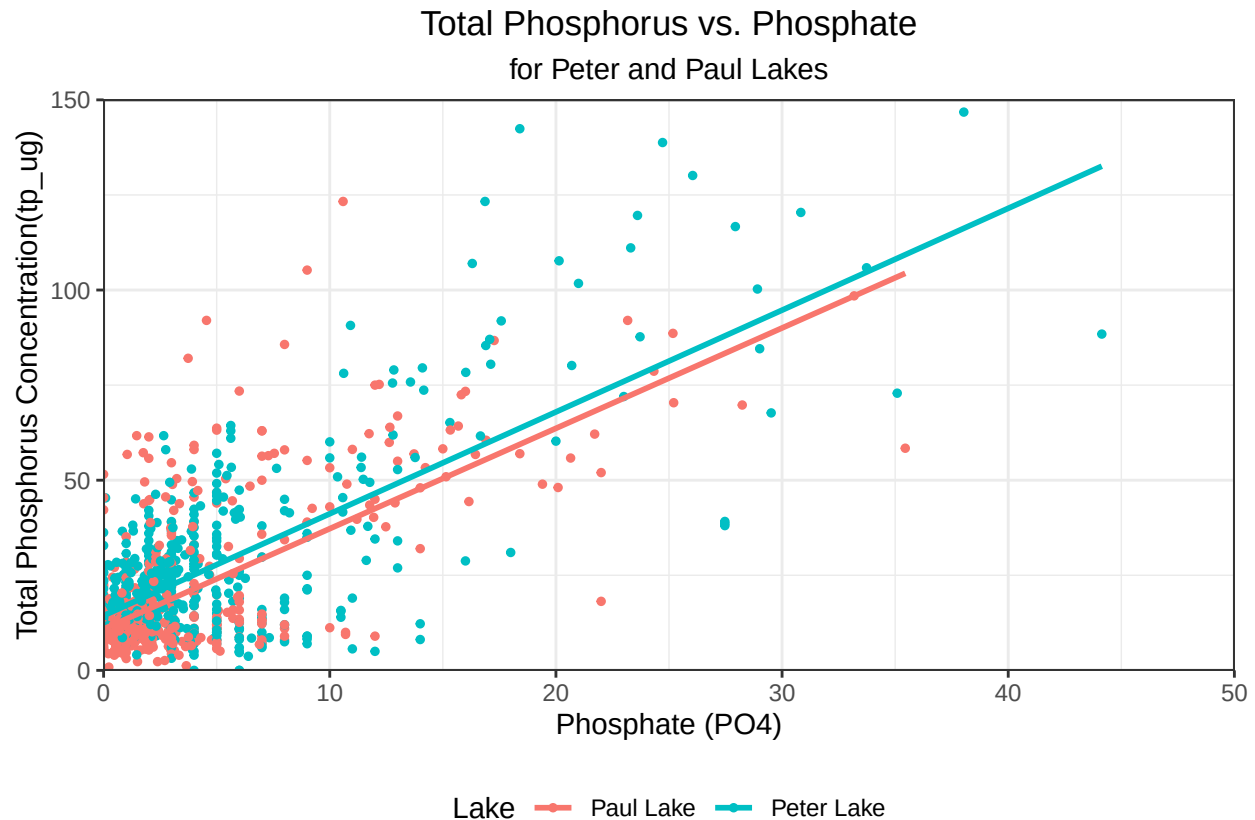
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 21948 rows containing non-finite outside the scale range
```

```
## ('stat_smooth()').
```

```
## Warning: Removed 21948 rows containing missing values or values outside the scale range
```

```
## ('geom_point()').
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5
CN$month = as.factor(CN$month)
boxplot_a = ggplot(CN, aes(x=month, y=temperature_C, color = lakename))+
  geom_boxplot()+
  labs(
    y = "Temperature (°C)",
    title = "Seasonal Water Quality"
  ) +
  theme_bw() +
  theme(
    axis.title.x = element_blank(),
    axis.text.x = element_blank(),
    plot.title = element_text(hjust = 0.5),
    legend.position = "none"
  )
```

```

boxplot_b = ggplot(CN, aes(x = month, y = tp_ug, color =
                        lakename)) +
  geom_boxplot() +
  labs(y = "Total Phosphorus (µg/L)", color = "Lake") +
  theme_bw() +
  theme(
    axis.title.x = element_blank(),
    axis.text.x = element_blank(),
    plot.title = element_text(hjust = 0.5)
  )

boxplot_c = ggplot(CN, aes(x = month, y = tn_ug, color =
                        lakename)) +
  geom_boxplot() +
  labs(
    x = "Month",
    y = "Total Nitrogen (µg/L)"
  ) +
  theme_bw() +
  theme(legend.position = "none")

plot_grid(
  boxplot_a,
  boxplot_b,
  boxplot_c,
  ncol = 1,
  align = 'v'
)

```

```

## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

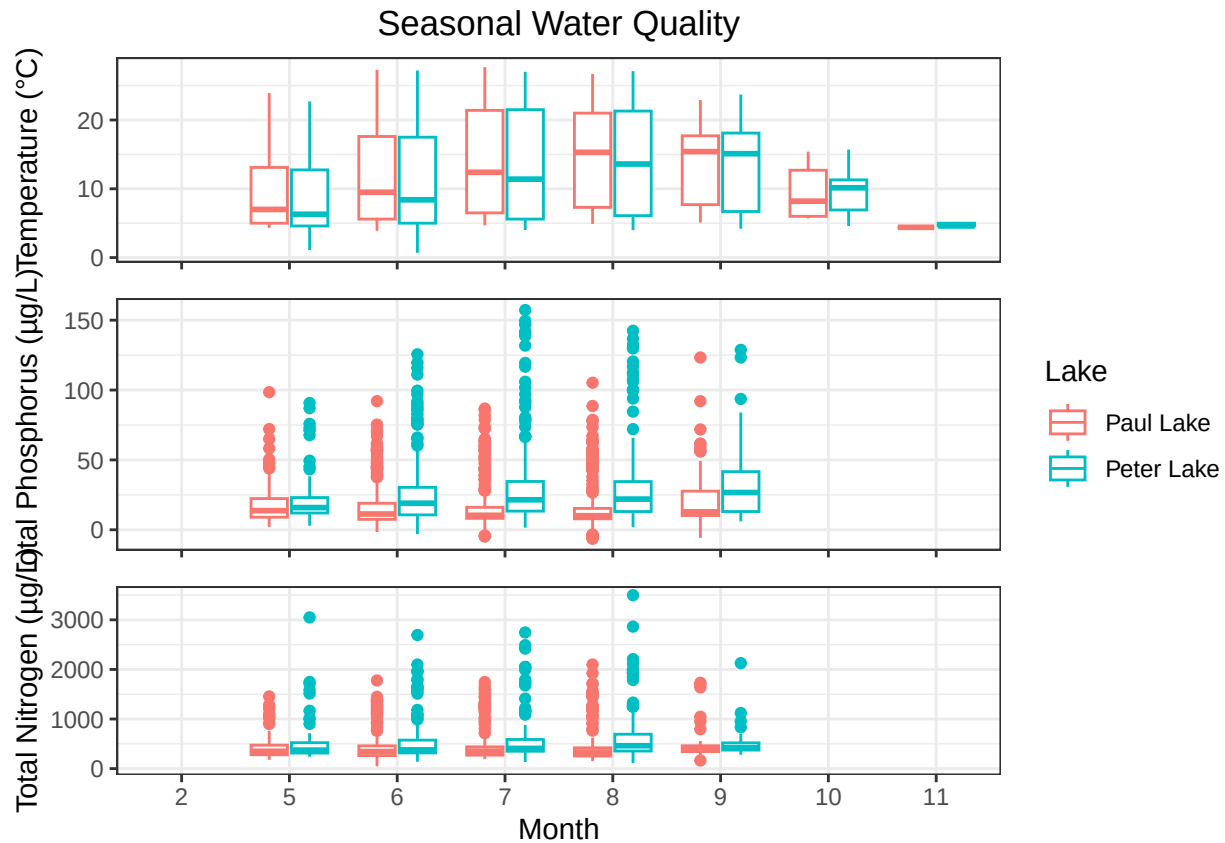
## Warning: Removed 20729 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```



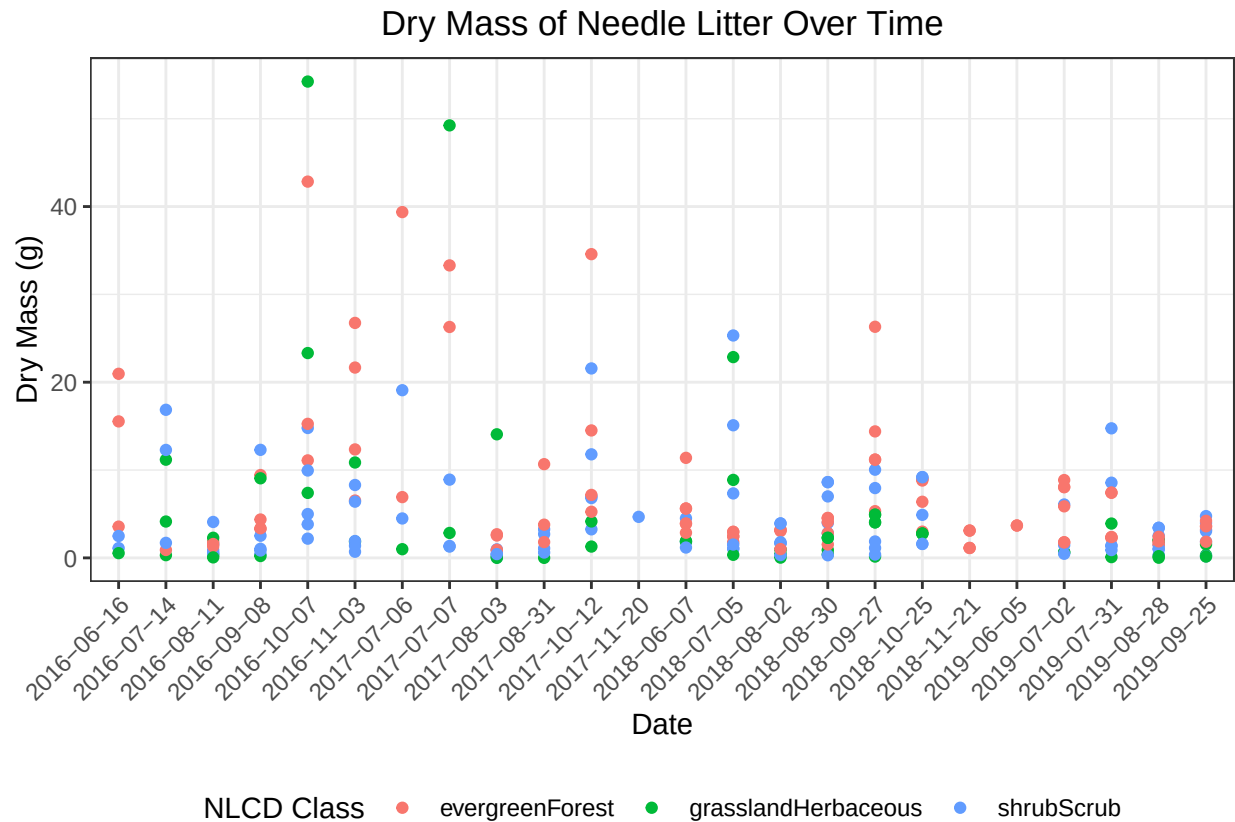
Question: What do you observe about the variables of interest over seasons and between lakes?

Answer: Both lakes share a nearly identical seasonal temperature pattern, Peter Lake has consistently and significantly higher concentrations of both phosphorus and nitrogen than Paul Lake.

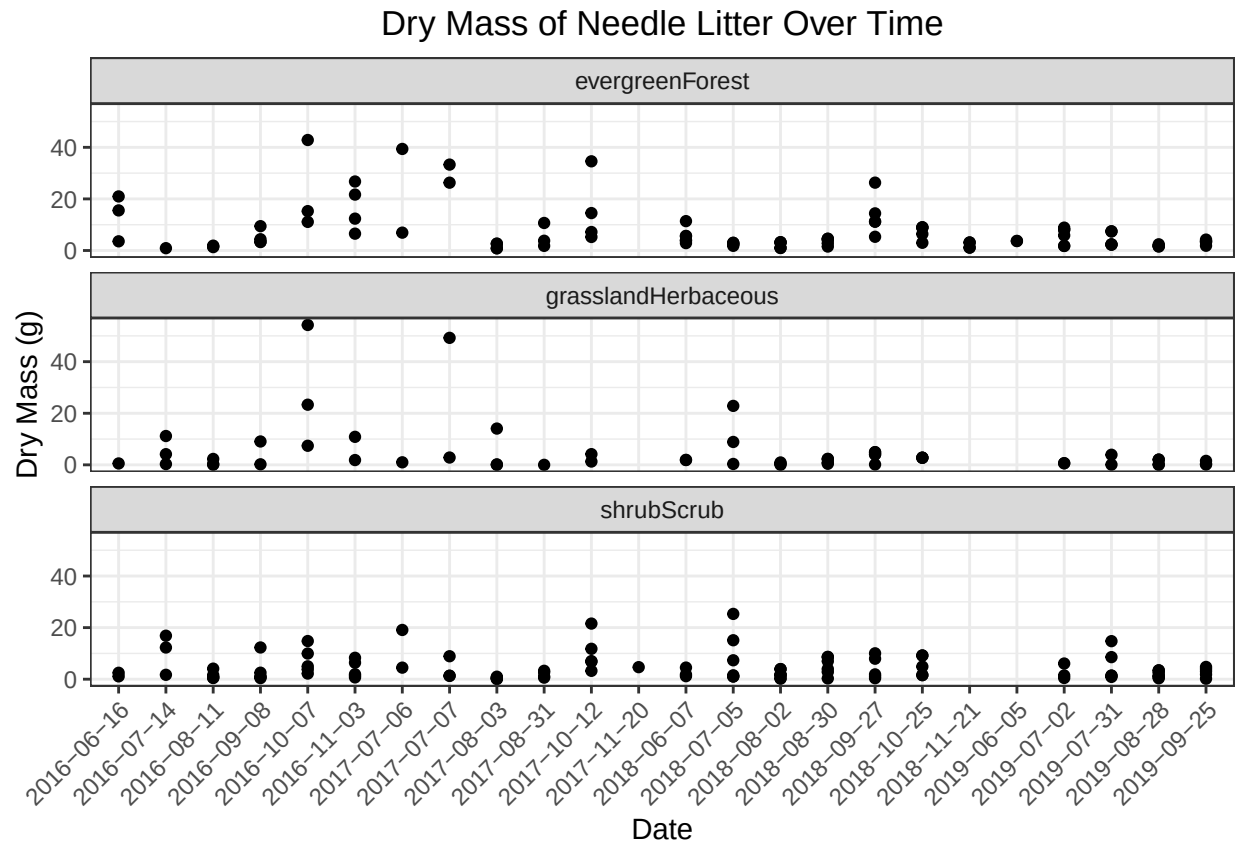
6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

#6

```
ggplot(subset(LMT, functionalGroup == "Needles" ),
  aes(x = collectDate, y = dryMass, color = nlcdClass)) +
  geom_point() +
  labs(
    title = "Dry Mass of Needle Litter Over Time",
    x = "Date",
    y = "Dry Mass (g)",
    color = "NLCD Class"
  ) +
  mytheme +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
#7
ggplot(subset(LMT, functionalGroup == "Needles" ),
       aes(x = collectDate, y = dryMass)) +
  geom_point() +
  facet_wrap(~ nlcdClass, ncol = 1) +
  labs(
    title = "Dry Mass of Needle Litter Over Time",
    x = "Date",
    y = "Dry Mass (g)"
  ) +
  mytheme +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: I think plot 7 is better because I can observe the trend for each cover class more clearly.